Simplifying expressions

You'll need to be able to **simplify** expressions that contain × and ÷ in your exam. Use these rules to help you.

1 Multiplying expressions

1. Multiply any number parts first.
2. Then multiply the letters. Remember to use $\square^2$ for letters which are multiplied twice or $\square^3$ for letters which are multiplied three times.

$10a \times 3a = 30a^2$

$10 \times 3 = 30$ $\quad a \times a = a^2$

$3s \times 6t = 18st$

$3 \times 6 = 18$ $\quad s \times t = st$

2 Dividing expressions

1. Write the division as a fraction.
2. Cancel any number parts.
3. If the same letter appears on the top and bottom, you can cancel that as well.

$8y \div 4 = \frac{{}^{2}\cancel{8}y}{\cancel{4}_{1}} = 2y$ $\quad 8 \div 4 = 2$

$\frac{{}^{4}\cancel{36}ab}{{}_{1}\cancel{9}b} = 4a$ $\quad 36 \div 9 = 4$

b appears on the top and the bottom, so cancel

Multiplying with algebra

You can multiply letters in algebra by writing them next to each other.

$ab = a \times b$

You can use indices to describe a letter multiplied by itself.

$y \times y = y^2$ You say 'y squared'.

You can use indices to describe the same letter multiplied together three times.

$n \times n \times n = n^3$ You say 'n cubed'.

For a reminder about squares and cubes have a look at page 8.

Worked example

(a) Simplify $a \times a \times a$ **(1 mark)**

a^3

(b) Simplify $7x \times 2y$ **(2 marks)**

$14xy$

(c) Simplify $10pq \div 2p$ **(2 marks)**

$\frac{{}^{5}\cancel{10}pq}{\cancel{2}p} = 5q$

(b) Multiply the numbers first and then the letters.
$7 \times 2 = 14$ $\quad x \times y = xy$

(c) Write the division as a fraction. You can cancel the number parts by dividing top and bottom by 2.
10 ÷ 2 = 5 so write 5 on top of your fraction.
p appears on the top and the bottom so you can cancel it. You are left with $5q$.

Now try this

Simplify

(a) $n \times n \times n \times n \times n$ **(1 mark)**

(b) $3r \times 2r \times r$ **(2 marks)**

(c) $4y \times 8z$ **(2 marks)**

(d) $20fg \div 5g$ **(2 marks)**

(e) $28wk \div 4wk$ **(2 marks)**

(d) $20fg \div 5g = \frac{20fg}{5g}$

Cancel any number parts then cancel any letters that are both on the top and bottom of the fraction.

Algebraic indices

You revised indices with numbers on page 9. You can also use the index laws to simplify powers in algebraic expressions.

y^3 — This part is called the **index**. The plural of index is indices. This part is called the **base**.

n^5 This means $n \times n \times n \times n \times n$.

Index laws

You can use these index laws to simplify powers and algebraic expressions.

1 To multiply powers of the same base, add the indices.

$a^m \times a^n = a^{m+n}$

$x^4 \times x^3 = x^{4+3} = x^7$

2 To divide powers of the same base, subtract the indices.

$a^m \div a^n = \frac{a^m}{a^n} = a^{m-n}$

$\frac{m^8}{m^2} = m^{8-2} = m^6$

3 To raise a power of a base to a further power, multiply the indices.

$(a^m)^n = a^{mn}$

$(j^2)^4 = j^{2\times 4} = j^8$

4 Powers apply to everything inside the brackets.

$(ab)^n = a^n \times b^n$

$(4b)^3 = 4^3 \times b^3 = 64b^3$

One at a time

When you are multiplying or dividing expressions with powers:

1. Multiply or divide any number parts first.
2. Use the index laws to work out the new power.

$7x \times 5x^6 = 35x^7$ — $x \times x^6 = x^{1+6} = x^7$

$7 \times 5 = 35$

$12 \div 3 = 4$

$\frac{12a^5}{3a^2} = 4a^3$

$a^5 \div a^2 = a^{5-2} = a^3$

(a) Add the indices. $a^m \times a^n = a^{m+n}$
(b) Subtract the indices. $a^m \div a^n = a^{m-n}$
(c) Multiply the indices. $(a^m)^n = a^{mn}$

Worked example

(a) Simplify $p^2 \times p^7$ **(1 mark)**

$p^2 \times p^7 = p^{2+7} = p^9$

(b) Simplify $m^8 \div m^3$ **(1 mark)**

$m^8 \div m^3 = m^{8-3} = m^5$

(c) Simplify $(a^6)^3$ **(1 mark)**

$(a^6)^3 = a^{6\times 3} = a^{18}$

Watch out!

You can only use the index laws when the bases are the same.

If there's no index then the number has a power of 1.

$n \times n^5 = n^{1+5} = n^6$

$x^8 \div x = x^{8-1} = x^7$

Now try this

1 (a) Simplify $m^5 \times m^4$ **(1 mark)**

(b) Simplify $k^6 \times k$ **(1 mark)**

2 (a) Simplify $h^{11} \div h^3$ **(1 mark)**

(b) Simplify $\frac{t^{25}}{t^5}$ **(1 mark)**

(c) Simplify $\frac{a^3 \times a^6}{a^4}$ **(1 mark)**

3 (a) Simplify $(y^4)^2$ **(1 mark)**

(b) Simplify $(5p^2)^3$ **(1 mark)**

Worked solution video

You could do this in two different ways:
$(5p^2)^3 = 5^3 \times (p^2)^3$
$(5p^2)^3 = 5p^2 \times 5p^2 \times 5p^2$
Both ways will give you the same answer.

Substitution

If you know the values of the letters in an algebraic expression, you can **substitute** them into the expression. This lets you work out the value of the expression.

$x = 7$ and $y = 2$ have been substituted into this expression.

$x + 5y$ — 5y means 5 × y. When $y = 2$, $5y = 5 \times 2 = 10$

$x = 7$ $\quad$ $y = 2$

$7 + 5 \times 2 = 7 + 10$ — Multiply before you Add.

$= 17$

When $x = 7$ and $y = 2$ the value of $x + 5y$ is 17.

Using BIDMAS

Remember to use the correct priority of operations when you are doing a calculation. BIDMAS gives the priority in which the operations should be carried out.

Brackets
Indices
Division
Multiplication
Addition
Subtraction

$2 \times 5^2 - (14 + 8)$
$= 2 \times 5^2 - 22$
$= 2 \times 25 - 22$
$= 50 - 22$
$= 28$

You should substitute all the values before doing any calculations.

Worked example

(a) Work out the value of $5x + 1$ when $x = -3$ **(2 marks)**

$5 \times (-3) + 1 = -15 + 1$
$= -14$

(b) Work out the value of $5p^3$ when $p = 2$ **(2 marks)**

$5 \times 2^3 = 5 \times 8$
$= 40$

(c) Work out the value of $3m + 4n$ when $m = 5$ and $n = -2$ **(2 marks)**

$3 \times 5 + 4 \times (-2) = 15 + (-8)$
$= 15 - 8$
$= 7$

(d) Work out the value of $2x(x - 1)$ when $x = 11$ **(2 marks)**

$2 \times 11 \times (11 - 1) = 2 \times 11 \times 10$
$= 22 \times 10$
$= 220$

Examiners' report

Be extra careful when substituting a negative number. You can use brackets around the number to make sure you don't make a mistake.

Real students have struggled with questions like this in recent exams – **be prepared!**

Remember BIDMAS. Indices comes before Multiplication.

Substitute all the values before starting your calculation.

You can multiply in any order.
$22 \times 10 = 2 \times 110$

Now try this

1 Work out the value of $h + k - 6n$ when $h = 5$, $k = 10$ and $n = \frac{1}{2}$ **(2 marks)**

2 Work out the value of $\frac{xw}{x + y}$ when $x = -8$, $w = 3$ and $y = 4$ **(3 marks)**

First work out what the top and bottom of the fraction come to, then work out the division.

3 $x = 3$, $y = 6$ and $z = 10$
Work out the value of these expressions.
(a) $x + z$ **(1 mark)**
(b) $yz - x^2$ **(1 mark)**
(c) $x^3 - (z - y)$ **(1 mark)**

Worked solution video